



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO MURUGAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences a tropical climate with monsoons from June to September. The route is prone to heavy rainfall during this period, leading to possible waterlogging and reduced visibility. In summer months (April-June), high temperatures can affect road conditions and vehicle performance.

3. Traffic Patterns and Congestion-Prone Areas: Traffic tends to peak in the mornings (7:30 AM to 10:00 AM) and late afternoons (5:00 PM to 7:30 PM), especially in towns like Sangagiri, Edappadi, and Omalur. The Salem - Kochi Hwy and major junctions such as NH44 and AH43 are known for congestion.

4. Assessment of Road Quality and Infrastructure: The road quality varies along this route. State highways are generally well-maintained, but some sections might have potholes or uneven surfaces. Pay attention to rural road segments near MDR851 where infrastructure may be less developed.

5. Suggestions for Alternative Routes for Emergencies: In emergencies, consider bypassing congested areas via the National Highway 44 (NH44), which runs parallel to parts of the route and offers better infrastructure and facilities.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials must adhere to national safety standards, such as proper labeling and documentation. Checkpoint clearances are mandatory in certain zones, especially in industrial areas.

7. Overview of Historical Incidents: There have been occasional reports of accidents involving heavy vehicles, typically due to poor road conditions during monsoons or driver fatigue. Recent data on hazardous materials incidents was minimal, indicating compliance and safety procedures are generally effective.

8. Environmental Considerations and Sensitive Areas: Drive responsibly near agricultural areas to prevent contamination. Regions near the Kaveri river require extra caution to prevent environmental impact from potential spills.

9. Communication Coverage and Dead Zones: Most of the route has adequate mobile network coverage. However, some rural stretches near K.N.Pudur might experience weak signals. Ensure communication devices are fully charged, and consider satellite phones for constant connectivity.

10. Estimated Emergency Response Times: Emergency services are generally accessible within 30-45 minutes in major towns. Response times may increase to over an hour in remote areas, so plan accordingly.

12. Overall Summary of Risk Assessment: This route presents moderate risk primarily due to varying road conditions and potential congestion. Weather can significantly impact travel, with monsoon rains being the primary concern. Proper pre-trip inspections, adherence to regulations, and preparation through alternative routes can mitigate most risks. Awareness of communication limitations and environmental impact considerations can further ensure a safe journey for vehicles transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45543, 77.81740	30 KM/Hr
8	Turn	High	11.45631, 77.81668	15 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	High	11.55608, 77.89762	15 KM/Hr
11	Turn	Medium	11.55640, 77.89913	30 KM/Hr
12	Turn	Medium	11.58249, 77.90829	30 KM/Hr
13	Turn	Medium	11.58294, 77.90775	30 KM/Hr
14	Turn	High	11.58320, 77.90646	15 KM/Hr
15	Turn	Medium	11.59290, 77.90736	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	Medium	11.67870, 77.96846	30 KM/Hr
21	Turn	Medium	11.70077, 77.97957	30 KM/Hr
22	Turn	High	11.70223, 77.97860	15 KM/Hr
23	Turn	Medium	11.71402, 77.99586	30 KM/Hr
24	Turn	Medium	11.71918, 78.01127	30 KM/Hr
25	Turn	Medium	11.72782, 78.02307	30 KM/Hr
26	Turn	Medium	11.73773, 78.03459	30 KM/Hr
27	Turn	Medium	11.73776, 78.03580	30 KM/Hr
28	Turn	Medium	11.73658, 78.03897	30 KM/Hr
29	Turn	Medium	11.74271, 78.04717	30 KM/Hr
30	Blind Spot	Blind Spot	11.87104, 78.08356	10 KM/Hr
31	Turn	Medium	11.87071, 78.08450	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
32	Turn	Medium	11.87089, 78.08558	30 KM/Hr
33	Turn	Medium	11.87249, 78.08707	30 KM/Hr
34	Turn	Medium	11.86893, 78.12084	30 KM/Hr
35	Turn	Medium	11.86491, 78.12535	30 KM/Hr
36	Turn	High	11.85834, 78.13562	15 KM/Hr
37	Turn	Medium	11.86602, 78.13898	30 KM/Hr
38	Turn	High	11.88415, 78.14479	15 KM/Hr
39	Turn	Medium	11.88479, 78.14463	30 KM/Hr
40	Turn	Medium	11.90585, 78.15291	30 KM/Hr
41	Turn	High	11.90765, 78.15205	15 KM/Hr
42	Turn	Medium	11.91236, 78.15597	30 KM/Hr
43	Turn	Medium	11.91313, 78.15617	30 KM/Hr
44	Turn	Medium	11.91520, 78.15840	30 KM/Hr
45	Turn	High	11.92455, 78.17076	15 KM/Hr
46	Turn	Medium	11.92820, 78.18042	30 KM/Hr
47	Turn	Medium	11.92795, 78.18173	30 KM/Hr
48	Turn	Medium	11.92901, 78.18346	30 KM/Hr
49	Turn	Medium	11.93572, 78.18713	30 KM/Hr
50	Turn	Medium	11.94907, 78.20171	30 KM/Hr
51	Turn	Medium	11.95120, 78.20653	30 KM/Hr
52	Turn	Medium	11.95200, 78.20703	30 KM/Hr
53	Turn	Medium	11.95257, 78.20853	30 KM/Hr
54	Turn	Medium	11.95187, 78.20963	30 KM/Hr
55	Turn	High	11.95220, 78.21182	15 KM/Hr
56	Turn	High	11.95160, 78.21285	15 KM/Hr
57	Turn	Medium	11.95232, 78.21421	30 KM/Hr
58	Turn	Medium	11.95223, 78.21525	30 KM/Hr
59	Turn	Medium	11.95295, 78.21620	30 KM/Hr
60	Turn	Medium	11.95263, 78.22474	30 KM/Hr
61	Turn	High	11.95178, 78.22604	15 KM/Hr
62	Turn	Medium	11.95847, 78.23410	30 KM/Hr
63	Turn	Medium	11.96196, 78.23438	30 KM/Hr
64	Turn	Medium	11.96509, 78.23292	30 KM/Hr
65	Turn	High	11.96667, 78.23277	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
66	Blind Spot	Blind Spot	11.96679, 78.23211	10 KM/Hr
67	Turn	High	11.97329, 78.23713	15 KM/Hr
68	Turn	High	11.97765, 78.23556	15 KM/Hr
69	Turn	High	11.98242, 78.24309	15 KM/Hr
70	Turn	High	11.98205, 78.24333	15 KM/Hr
71	Turn	High	11.98346, 78.24637	15 KM/Hr
72	Turn	High	11.96949, 78.26940	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
2	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
3	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
4	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
28	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
29	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
30	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
31	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
32	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
33	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium
34	hospital	Raja Hospital, Dharmapuri	11.9792908, 78.2509482	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55608, 77.89762



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58320, 77.90646



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59290, 77.90736



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70077, 77.97957



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70223, 77.97860



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71402, 77.99586



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71918, 78.01127



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72782, 78.02307



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73773, 78.03459



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73776, 78.03580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73658, 78.03897



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74271, 78.04717



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.87104, 78.08356



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87071, 78.08450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87089, 78.08558



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87249, 78.08707

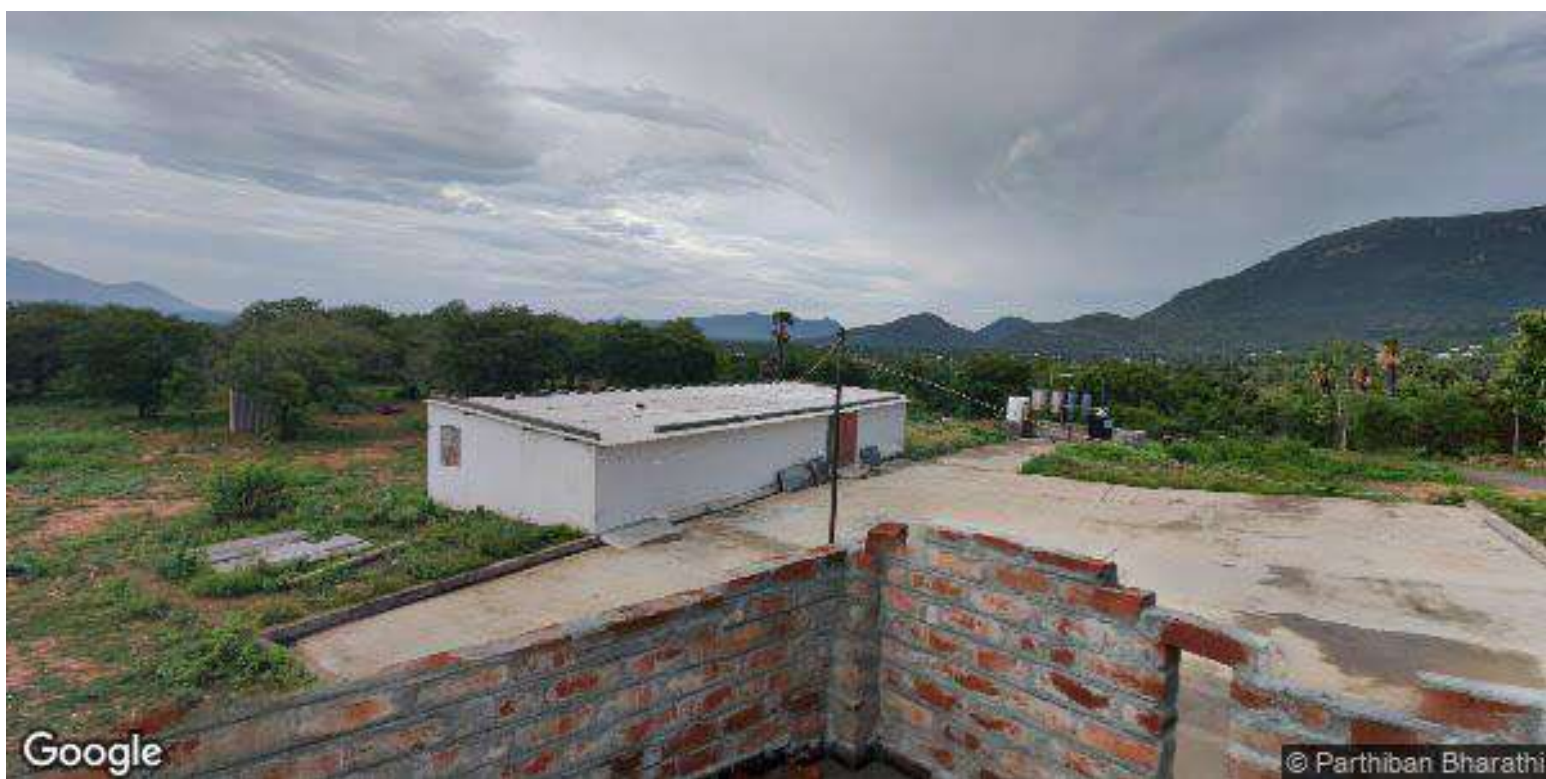


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86893, 78.12084



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95120, 78.20653



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97329, 78.23713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97765, 78.23556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98242, 78.24309



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98205, 78.24333



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98346, 78.24637



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96949, 78.26940
